

Topic 18A: Arenes - benzene

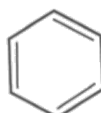
1. understand that the bonding in benzene has been represented using the Kekulé and the delocalised model, the latter in terms of overlap of p -orbitals to form π -bonds

2. understand that evidence for the delocalised model of the bonding in benzene is provided by data from enthalpy changes of hydrogenation and carbon-carbon bond lengths

Students may represent the structure of benzene as:



or



as appropriate in equations and mechanisms.

3. understand why benzene is resistant to bromination, compared with alkenes, in terms of delocalisation of π -bonds in benzene and the localised electron density of the π -bond in alkenes

4. understand the reactions of benzene with:

- i oxygen in air (combustion with a smoky flame)
- ii bromine, in the presence of a catalyst
- iii a mixture of concentrated nitric and sulfuric acids
- iv halogenoalkanes and acyl chlorides with aluminium chloride as catalyst (Friedel-Crafts reaction)

5. understand the mechanism of the electrophilic substitution reactions of benzene (halogenation, nitration and Friedel-Crafts reactions), including the generation of the electrophile

6. understand the reaction of phenol with bromine water